

ADVANCED MATRIX ALGEBRA & MATHEMATICAL ANALYSIS

Volume III: Expanded Theoretical Proofs, Determinant Identities, and Deep Solution Matrices

A Continuous Formal Text for Engineering, Physical Sciences, and Advanced Linear Computing without LaTeX Elements

VOLUME CORE ANALYTICAL THEMES

This textbook volume contains full formal derivations, rigorous geometric interpretation of transformations, advanced property logs, and high-density multi-stage problem matrix sheets mapping all algebraic properties of linear operators.

Chapter Domain	Detailed Analytical Core Components
1. Foundations & Element Algebra	Double-subscript field theory, structural dimensions, Kronecker delta applications in identity mapping.
2. Advanced Properties of Operations	Commutator brackets, transpose identities, trace of a matrix, orthogonally transformed states.
3. Determinants & Permutations	Sarrus rule restrictions, full Laplacian expansions, sign tensor fields, algebraic minors.
4. Rigorous Axioms & Proof Blocks	Row reduction invariance, multiplicative theorem verification, row singularity behavior.
5. Classical Adjoint Mechanics	Cofactor matrix grids, orthogonal transposes, adjugate polynomial identities.
6. Full Inversion & Invertibility	Singularity proofs, Reversal law properties, computational blocks via Cramer system linkages.
7. Dense Advanced Numerical Problems	High-dimension matrix algebra, full arithmetic steps, completely expanded proofs.

1. Foundational Matrix Algebra and Element Field Mapping

1.1 Field Array Mapping Definitions

An $m \times n$ matrix is a structured layout of elements map-positioned across m row vectors and n column vectors over a chosen number field (real or complex numbers). Using standard compact notation, the entire matrix structure can be defined by enclosing its individual elements in brackets:

$$A = [a_{ij}] \text{ where } 1 \leq i \leq m \text{ and } 1 \leq j \leq n$$

The index i represents the exact row lane, while j maps the column column track. The mathematical properties of the identity matrix can be elegantly defined using the **Kronecker Delta** function (δ_{ij}), which serves as a filter during index matching operations:

$$\delta_{ij} = 1 \text{ if } i = j; \text{ and } \delta_{ij} = 0 \text{ if } i \neq j$$

Using this delta filter, an identity matrix I is simply a square matrix where every entry is defined by $a_{ij} = \delta_{ij}$.

1.2 Transposition Theory

Transposing a matrix flips its structure across its main diagonal, turning all rows into columns and all columns into rows. If the original matrix A has dimensions of $m \times n$, its transpose A^T will have dimensions of $n \times m$. The individual elements shift positions according to the following rule:

$$[a^T]_{ij} = a_{ji}$$

Transposition operations follow several essential algebraic properties across matrix sums and products:

- **Double Transpose Law:** Transposing a matrix twice returns it to its original state: $(A^T)^T = A$
- **Additive Distribution:** Transposition distributes evenly across matrix addition: $(A + B)^T = A^T + B^T$
- **Scalar Factoring:** Multiplying by a constant scalar commutes with transposition: $(kA)^T = k(A^T)$

- **The Reversal Law for Transposes:** Transposing the product of two matrices reverses their multiplication order: $(AB)^T = B^T \times A^T$

2. Advanced Operational Properties & Trace Algebra

2.1 Matrix Multiplication and Non-Commutativity Bounds

While standard number algebra is always commutative, matrix operations generally do not commute. For any two square matrices $\mathbf{A}$ and $\mathbf{B}$, the products $\mathbf{AB}$ and $\mathbf{BA}$ are rarely identical. To measure this structural difference, we calculate the **Commutator Bracket**, which is defined as:

$$[\mathbf{A}, \mathbf{B}] = \mathbf{AB} - \mathbf{BA}$$

If two linear operator matrices commute perfectly, their commutator bracket equals the zero matrix ($[\mathbf{A}, \mathbf{B}] = \mathbf{O}$). If they do not commute, $\mathbf{AB} \neq \mathbf{BA}$, meaning the order of operations must be strictly maintained.

2.2 The Trace Function

The **Trace** of a square matrix is a single scalar value calculated by summing all the elements along its main diagonal. The trace function is denoted as $\text{tr}(\mathbf{A})$:

$$\text{tr}(\mathbf{A}) = \sum_{i=1 \text{ to } n} a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$$

The trace function possesses unique and highly useful algebraic properties:

1. **Linearity:** The trace handles addition and scalar multiplication linearly: $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$ and $\text{tr}(k\mathbf{A}) = k \times \text{tr}(\mathbf{A})$.
2. **Transposition Invariance:** Transposing a matrix does not alter its diagonal entries, so the trace remains unchanged: $\text{tr}(\mathbf{A}^T) = \text{tr}(\mathbf{A})$.
3. **Cyclic Property:** The trace of a matrix product is invariant under cyclic shifts of the multiplication order, even though the resulting matrices themselves are different:

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

Extending this rule to three matrices yields: $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}) = \text{tr}(\mathbf{CAB})$. Note that non-cyclic permutations, like $\text{tr}(\mathbf{BAC})$, are generally not equal to $\text{tr}(\mathbf{ABC})$.

3. Comprehensive Determinant Theory & Laplacian Decompositions

3.1 Permutations and Sign Tensors

The determinant of an $n \times n$ square matrix is a single scalar value that summarizes its geometric scaling properties. Formally, it can be expressed as a sum over all possible index permutations:

$$\det(A) = \sum_{(p \text{ elements of } S_n)} \text{sgn}(p) \times a_{1p(1)} \times a_{2p(2)} \times \dots \times a_{np(n)}$$

The function **sgn(p)** acts as a sign filter, returning **+1** if a permutation requires an even number of swaps, and **-1** if it requires an odd number of swaps.

3.2 Minors and Cofactors

For larger matrices, calculating the determinant directly from permutations becomes inefficient. Instead, we use minors and cofactors to break the matrix down into smaller sub-matrices:

- **Minor (M_{ij}):** The determinant of the smaller sub-matrix that remains after crossing out the i -th row and the j -th column from the original square matrix.
- **Cofactor (C_{ij}):** The minor value multiplied by a positional sign factor based on its coordinates in the grid:

$$C_{ij} = (-1)^{i+j} \times M_{ij}$$

3.3 General Laplacian Expansion Theorem

The Laplacian expansion theorem states that the determinant of any square matrix can be calculated by expanding along any single row or column. The total determinant is the sum of the elements in that line multiplied by their corresponding cofactors:

- **Expansion across Row i :**

$$\det(A) = \sum_{j=1 \text{ to } n} (a_{ij} \times C_{ij}) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}$$

• **Expansion down Column j:**

$$\det(A) = \sum_{i=1 \text{ to } n} (a_{ij} \times C_{ij}) = a_{1j}C_{11} + a_{2j}C_{21} + \dots + a_{nj}C_{nj}$$

4. Complete Axiomatic Analysis of Determinant Identities

Using determinant properties allows us to simplify complex matrices and perform efficient calculations without expanding every cofactor. The fundamental rules are detailed below:

1. **Row/Column Interchanging (Sign Flips):** Swapping any two rows or any two columns multiplies the total determinant by -1 .
2. **Scalar Factoring Mechanics:** Factoring a scalar constant k out of a single row or column scales the whole determinant by k . Consequently, pulling a scalar out of an entire $n \times n$ matrix scales the determinant exponentially:

$$\det(kA) = k^n \times \det(A)$$

3. **Linear Invariance under Row Additions:** Adding a scalar multiple of one row to another row does not alter the determinant's value. This property allows us to use Gaussian row reduction to transform a matrix into an upper triangular form, where the determinant is simply the product of the diagonal elements:

$$\det(A) = a_{11} \times a_{22} \times \dots \times a_{nn}$$

4. **Multiplicative Theorem:** The determinant of a product of two square matrices equals the product of their individual determinants:

$$\det(AB) = \det(A) \times \det(B)$$

5. **Singularity Criteria:** A matrix is singular (cannot be inverted) if and only if its determinant is exactly zero ($\det(A) = 0$). This occurs whenever a matrix contains rows or columns that are identical, proportional, or all zeros.

5. Classical Adjoint Mechanics & Cofactor Formulations

5.1 The Adjugate Mapping Model

The **Adjoint Matrix** (or adjugate matrix) is constructed by calculating the cofactors for all elements in a square matrix and then transposing that cofactor grid. It is denoted as ***adj(A)***:

$$\text{adj}(A) = [C_{ij}]^T = [C_{ji}]$$

This means the element at row ***i*** and column ***j*** in the adjoint matrix is equal to the cofactor calculated from row ***j*** and column ***i*** of the original matrix.

5.2 The Fundamental Adjoint Identity

Multiplying a square matrix by its adjoint always results in a scalar matrix where the diagonal entries equal the matrix's determinant. This property holds true regardless of the multiplication order:

$$A \times \text{adj}(A) = \text{adj}(A) \times A = \det(A) \times I$$

Where ***I*** represents the identity matrix of the same dimensions.

5.3 Essential Adjoint Properties

The adjoint matrix follows several vital identities that are useful when simplifying complex algebraic expressions:

- **Adjoint of a Product:** The adjoint of a product of two matrices equals the product of their individual adjoints taken in reverse order: ***adj(AB) = adj(B) × adj(A)***
- **Adjoint of a Transpose:** Transposing and finding the adjoint commute perfectly: ***adj(A^T) = (adj(A))^T***
- **Determinant of an Adjoint:** The determinant of an adjoint matrix scales polynomially based on the original matrix's order ***n***:

$$\det(\text{adj}(A)) = (\det(A))^{n-1}$$

6. Full Inversion & Invertibility Analysis

6.1 Analytical Inversion Model

The inverse of a square matrix A is another unique matrix, denoted as A^{-1} , that satisfies the following property:

$$A \times A^{-1} = A^{-1} \times A = I$$

By utilizing the fundamental adjoint identity ($A \times \text{adj}(A) = \det(A) \times I$), we can derive the analytical formula for the inverse matrix. If the matrix is non-singular ($\det(A) \neq 0$), dividing both sides by the scalar determinant yields:

$$A^{-1} = (1 / \det(A)) \times \text{adj}(A)$$

6.2 Key Inversion Identities

Matrix inversion operations adhere to several rigid algebraic laws:

1. **The Reversal Law for Inverses:** Inverting a product of multiple matrices reverses their multiplication order:

$$(AB)^{-1} = B^{-1} \times A^{-1}$$

2. **Determinant Reciprocity:** The determinant of an inverse matrix is the mathematical reciprocal of the original matrix's determinant:

$$\det(A^{-1}) = 1 / \det(A)$$

3. **Adjoint Inversion Law:** The inverse of an adjoint matrix matches the adjoint of the inverse matrix:

$$(\text{adj}(A))^{-1} = \text{adj}(A^{-1}) = (1 / \det(A)) \times A$$

7. High-Density Advanced Numerical Problem Sets

PROBLEM 1: COMPLEX 3X3 MATRIX MULTIPLICATION & TRACE VERIFICATION

Given the two 3x3 square matrices A and B :

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 0 & 4 \\ -1 & 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 2 & 1 \\ 1 & 4 & 3 \end{bmatrix}$$

Compute: (a) the product matrix AB , (b) the product matrix BA , and (c) verify that $\text{tr}(AB) = \text{tr}(BA)$.

SOLUTION:

Step 1: Calculate the product matrix AB using row-by-column dot products.

- Row 1 entries:

$$c_{11} = (1 \times 2) + (-2 \times -3) + (3 \times 1) = 2 + 6 + 3 = 11$$

$$c_{12} = (1 \times 1) + (-2 \times 2) + (3 \times 4) = 1 - 4 + 12 = 9$$

$$c_{13} = (1 \times 0) + (-2 \times 1) + (3 \times 3) = 0 - 2 + 9 = 7$$

- Row 2 entries:

$$c_{21} = (2 \times 2) + (0 \times -3) + (4 \times 1) = 4 + 0 + 4 = 8$$

$$c_{22} = (2 \times 1) + (0 \times 2) + (4 \times 4) = 2 + 0 + 16 = 18$$

$$c_{23} = (2 \times 0) + (0 \times 1) + (4 \times 3) = 0 + 0 + 12 = 12$$

- Row 3 entries:

$$c_{31} = (-1 \times 2) + (3 \times -3) + (1 \times 1) = -2 - 9 + 1 = -10$$

$$c_{32} = (-1 \times 1) + (3 \times 2) + (1 \times 4) = -1 + 6 + 4 = 9$$

$$c_{33} = (-1 \times 0) + (3 \times 1) + (1 \times 3) = 0 + 3 + 3 = 6$$

$$AB = \begin{bmatrix} 11 & 9 & 7 \\ 8 & 18 & 12 \\ -10 & 9 & 6 \end{bmatrix}$$

Step 2: Calculate the product matrix BA .

- Row 1 entries:

$$d_{11} = (2 \times 1) + (1 \times 2) + (0 \times -1) = 2 + 2 + 0 = 4$$

$$d_{12} = (2 \times -2) + (1 \times 0) + (0 \times 3) = -4 + 0 + 0 = -4$$

$$d_{13} = (2 \times 3) + (1 \times 4) + (0 \times 1) = 6 + 4 + 0 = 10$$

- Row 2 entries:

$$d_{21} = (-3 \times 1) + (2 \times 2) + (1 \times -1) = -3 + 4 - 1 = 0$$

$$d_{22} = (-3 \times -2) + (2 \times 0) + (1 \times 3) = 6 + 0 + 3 = 9$$

$$d_{23} = (-3 \times 3) + (2 \times 4) + (1 \times 1) = -9 + 8 + 1 = 0$$

- Row 3 entries:

$$d_{31} = (1 \times 1) + (4 \times 2) + (3 \times -1) = 1 + 8 - 3 = 6$$

$$d_{32} = (1 \times -2) + (4 \times 0) + (3 \times 3) = -2 + 0 + 9 = 7$$

$$d_{33} = (1 \times 3) + (4 \times 4) + (3 \times 1) = 3 + 16 + 3 = 22$$

$$BA = \begin{vmatrix} 4 & -4 & 10 \\ 0 & 9 & 0 \\ 6 & 7 & 22 \end{vmatrix}$$

Step 3: Calculate and verify the traces.

Sum the main diagonal entries for both product matrices:

$$\text{tr}(AB) = 11 + 18 + 6 = 35$$

$$\text{tr}(BA) = 4 + 9 + 22 = 35$$

The calculation confirms $\text{tr}(AB) = \text{tr}(BA) = 35$, validating the cyclic trace property.

PROBLEM 2: COMPLETE ANALYTICAL INVERSION OF A 3X3 GRID WITH FRACTION FIELDS

Given the 3x3 square matrix A :

$$A = \begin{vmatrix} 2 & -1 & 1 \\ 1 & 3 & 2 \\ 3 & 1 & 4 \end{vmatrix}$$

Perform a complete cofactor decomposition to calculate its determinant, adjoint, and unique inverse matrix (A^{-1}).

SOLUTION:

Step 1: Calculate the determinant of matrix A .

Expanding across the first row yields:

$$|A| = 2 \times ((3 \times 4) - (2 \times 1)) - (-1) \times ((1 \times 4) - (2 \times 3)) + 1 \times ((1 \times 1) - (3 \times 3))$$

$$|A| = 2 \times (12 - 2) + 1 \times (4 - 6) + 1 \times (1 - 9)$$

$$|A| = 2 \times (10) + 1 \times (-2) + 1 \times (-8) = 20 - 2 - 8 = 10$$

Since $|A| = 10 \neq 0$, the matrix is non-singular and invertible.

Step 2: Calculate all nine individual cofactors.

$$\bullet C_{11} = +((3 \times 4) - (2 \times 1)) = 12 - 2 = 10$$

$$\bullet C_{12} = -((1 \times 4) - (2 \times 3)) = -(4 - 6) = 2$$

$$\bullet C_{13} = +((1 \times 1) - (3 \times 3)) = 1 - 9 = -8$$

$$\bullet C_{21} = -((-1 \times 4) - (1 \times 1)) = -(-4 - 1) = 5$$

$$\bullet C_{22} = +((2 \times 4) - (1 \times 3)) = 8 - 3 = 5$$

$$\bullet C_{23} = -((2 \times 1) - (-1 \times 3)) = -(2 + 3) = -5$$

$$\bullet C_{31} = +((-1 \times 2) - (1 \times 3)) = -2 - 3 = -5$$

$$\bullet C_{32} = -((2 \times 2) - (1 \times 1)) = -(4 - 1) = -3$$

$$\bullet C_{33} = +((2 \times 3) - (-1 \times 1)) = 6 + 1 = 7$$

Step 3: Assemble the Cofactor Matrix and transpose it to find the Adjoint Matrix.

Arrange the cofactors into a grid, then swap its rows and columns:

$$\text{adj}(A) = \begin{vmatrix} 10 & 5 & -5 \\ 2 & 5 & -3 \\ -8 & -5 & 7 \end{vmatrix}$$

Step 4: Compute the final inverse matrix $A^{-1} = (1 / |A|) * \text{adj}(A)$.

Divide every entry in the adjoint matrix by the scalar determinant value (10):

$$A^{-1} = \begin{vmatrix} 10/10 & 5/10 & -5/10 \\ 2/10 & 5/10 & -3/10 \\ -8/10 & -5/10 & 7/10 \end{vmatrix} = \begin{vmatrix} 1.0 & 0.5 & -0.5 \\ 0.2 & 0.5 & -0.3 \\ -0.8 & -0.5 & 0.7 \end{vmatrix}$$

PROBLEM 3: FINDING THE DETERMINANT OF A 4X4 MATRIX USING GAUSSIAN ROW REDUCTION

Calculate the determinant of the following 4x4 square matrix D :

$$D = \begin{vmatrix} 1 & 2 & -1 & 0 \\ 2 & 3 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 4 \end{vmatrix}$$

SOLUTION:

Step 1: Perform row operations to create zeros under the first element of Column 1.

Row 1 is our pivot row. We want to clear out the entries in row 2 and row 4 of column 1:

- Operation: $R_2 \leftarrow R_2 - 2R_1$
New Row 2 = $[2-2(1), 3-2(2), 1-2(-1), 1-2(0)] = [0, -1, 3, 1]$
- Operation: $R_4 \leftarrow R_4 - R_1$
New Row 4 = $[1-1, 0-2, 1-(-1), 4-0] = [0, -2, 2, 4]$

After these steps, the updated matrix looks like this:

$$\begin{vmatrix} 1 & 2 & -1 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & -2 & 2 & 4 \end{vmatrix}$$

Step 2: Create zeros under the pivot element of Column 2.

Using the updated Row 2 as our pivot row, we clear the entries below it in column 2:

- Operation: $R_3 \leftarrow R_3 + R_2$
New Row 3 = $[0, 1+(-1), 2+3, 3+1] = [0, 0, 5, 4]$
- Operation: $R_4 \leftarrow R_4 - 2R_2$
New Row 4 = $[0, -2-2(-1), 2-2(3), 4-2(1)] = [0, 0, -4, 2]$

The matrix is now transformed into this state:

$$\begin{vmatrix} 1 & 2 & -1 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & -4 & 2 \end{vmatrix}$$

Step 3: Clear the final entry in Column 3 to achieve an upper triangular matrix.

We use Row 3 to eliminate the value in row 4 of column 3. To simplify the arithmetic, add a scaled version of row 3:

Operation: $R_4 \leftarrow R_4 + 0.8R_3$

New Row 4 = $[0, 0, -4 + 0.8(5), 2 + 0.8(4)] = [0, 0, 0, 2 + 3.2] = [0, 0, 0, 5.2]$

The matrix is now fully converted into an upper triangular form:

$$\begin{vmatrix} 1 & 2 & -1 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 5.2 \end{vmatrix}$$

Step 4: Calculate the determinant by multiplying the diagonal entries.

Since row operations do not alter the determinant's value, we find the total determinant by multiplying down the main diagonal:

$$\det(D) = 1 \times (-1) \times 5 \times 5.2 = -5 \times 5.2 = -26$$

Conclusion: The determinant of the 4x4 matrix D is exactly -26.